

NAUBINWAY, MI, USA

WMO: 997793

Lat: 46.0869N

Lon: 85.4444W

Elev: 591

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.9	-2.6	N/A	N/A	N/A	N/A	N/A	N/A	32.2	22.1	27.7	24.7	5.5	300	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	76.0	N/A	74.0	N/A	72.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.4	24.7	20.9	DB	-14.9	81.8	7.1	2.6	-20.0	83.7	-24.1	85.2	-28.1	86.7	-33.2	88.6	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	17.5	16.8	25.6	37.8	49.3	58.8	64.3	64.4	58.5	47.5	36.0	25.1	(6)	
	DBStd	18.63	10.64	10.07	10.10	6.78	6.41	5.51	4.92	4.47	6.09	7.03	8.20	8.72	(7)	
	HDD50	4487	1007	929	755	369	90	4	0	0	6	134	422	771	(8)	
	HDD65	8543	1472	1349	1220	817	489	197	74	66	209	544	870	1236	(9)	
	CDD50	1544	0	0	0	2	67	268	443	446	260	56	2	0	(10)	
	CDD65	123	0	0	0	0	1	10	52	47	13	0	0	0	(11)	
	CDH74	180	0	0	0	0	3	23	90	58	6	0	0	0	(12)	
	CDH80	11	0	0	0	0	0	1	8	2	0	0	0	0	(13)	
(14) Wind	WSAvg	9.4	9.9	9.4	9.5	10.3	9.3	8.1	7.4	8.0	9.3	11.0	11.0	10.1	(14)	
(15) Precipitation	PrecAvg	33.4	2.2	1.6	2.2	2.6	2.9	3.1	3.0	3.0	4.2	3.4	3.0	2.1	(15)	
	PrecMax	44.6	4.6	3.7	6.8	5.6	4.8	7.0	5.0	5.0	8.5	6.4	4.5	5.1	(16)	
	PrecMin	24.0	1.0	0.4	0.1	0.8	0.7	1.2	1.1	1.1	1.0	1.3	1.2	0.1	(17)	
	PrecStd	4.8	0.9	0.8	1.5	1.3	1.1	1.3	1.2	1.1	1.7	1.5	0.9	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	37.7	53.8	56.8	71.8	77.5	80.4	78.3	74.7	66.5	55.1	45.7	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	36.2	34.8	46.7	53.0	65.7	72.7	76.2	75.4	71.3	63.6	51.6	42.1	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	34.5	33.3	42.0	50.2	62.1	69.2	74.2	73.7	69.5	61.0	49.4	39.6	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	32.7	31.6	38.1	47.5	58.8	66.6	72.2	71.9	67.9	58.8	47.6	36.5	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	MDBR		12.5	14.6	14.7	11.8	12.6	12.2	12.0	12.2	12.0	10.9	9.6	10.3	(35)	
	5% DB	MCDBR	9.3	11.8	14.4	15.4	16.3	15.8	14.1	13.2	12.1	11.4	9.7	11.1	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.283	0.288	0.313	0.354	0.376	0.393	0.391	0.380	0.367	0.335	0.303	0.275	(40)	
	taud		2.412	2.383	2.404	2.352	2.362	2.348	2.373	2.405	2.432	2.503	2.521	2.485	(41)	
	Ebn at Noon		262	284	292	286	282	277	276	275	269	263	253	254	(42)	
	Edh at Noon		22	28	32	37	38	38	37	35	31	25	20	19	(43)	
(44) All-Sky Solar Radiation	RadAvg		350	643	1068	1435	1777	1922	1933	1638	1232	717	376	266	(44)	
	RadStd		38	67	116	157	147	134	93	105	96	83	51	25	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page